

company opts for verified, high-quality removal mechanisms with a long-term impact rather than just compliance offsets. Despite all this, the road to 2030 seems to be a tough one for Microsoft, as rapid AI and data centre expansions have driven total emissions up compared to a 2020 baseline. This is a common challenge that most of the big tech firms, including Google and Meta, are facing now.

Amazon: One of Amazon's most significant contributions to the global net-zero campaign is The Climate Pledge that it co-founded in 2019. Through this, it has mobilised more than 500 companies to join its journey to reach net-zero emissions by 2040. Amazon's net-zero strategy involves heavy investment in carbon-free energy, including nuclear and solar, matching 100% of its electricity consumption with renewable sources, and electrifying its delivery fleet with over 100,000 electric vans and e-bikes. It also works with suppliers to reduce scope 3 emissions. To address emissions that cannot be eliminated, Amazon invests in high-integrity carbon credits and nature-



Replanting saplings in areas like the Amazon, where the forests have dwindled, is a sure-shot way to offset carbon emissions

based solutions like forest conservation through initiatives such as the LEAF Coalition. The company also has its own carbon credits marketplace, called the Amazon Sustainability Exchange, and supports the development of sustainability technologies through the Climate Pledge Fund.

Google: Google aims to achieve net-zero emissions across all their operations and value chain by 2030. Despite a 27% year-on-year increase in electricity consumption by their data centres due to the AI boom, in 2024 they managed to reduce data centre

energy emissions by 12% compared to 2023. They are one of the largest corporate purchasers of clean energy, and have matched 100% of their global electricity use with renewable energy purchases since 2017. Like other majors, they are also committed to water consumption and waste management,

advocating a circular economy. Like Apple, they have also been busy innovating on the product front, building consumer devices with recycled materials, eliminating plastic in their packaging, and making it easy for customers to repair, reuse, and recycle their devices. The products they launched and manufactured in 2024 had at least 20% recycled content, including magnets and aluminium enclosures.

One of Google's most interesting investments in nature-based carbon removal is with Brazilian startup Mombak, which reforests degraded pasturelands in the Amazon using more than 80 native and biodiverse tree species, including a few endangered ones. To scientifically quantify the increase in biodiversity, Google and Mombak use Google DeepMind's Perch AI model to analyse sound recordings of birds and insects from the restored areas. Google has also invested in an Indian company called Varaha, which converts agricultural waste into biochar, a charcoal-like substance that sequesters carbon and improves soil health.

Other major tech companies like Meta, Tesla, Intel, Lenovo, and Salesforce are also following similar approaches to achieve their ambitious net-zero emission targets.

Indian tech majors

- Infosys became carbon neutral in 2020 using a combination

India's Carbon Credit Trading System

- India's Carbon Credit Trading Scheme (CCTS), which lays the foundation for the Indian Carbon Market (ICM), was introduced under the Energy Conservation (Amendment) Act, 2022, and is expected to start compliance trading from mid-2026.
- The unit of trade in the ICM is a Carbon Credit Certificate (CCC), which is equal to one metric tonne of carbon dioxide.
- India has opted for a baseline-and-credit approach to carbon credit trading.
- The CCTS is managed by the Bureau of Energy Efficiency (BEE), which sets the baselines, verifies credits, and issues CCCs.
- The BEE assigns emission reduction targets to obligated entities (in energy-intensive industries), under the guidance of a national steering committee. Entities that outperform the targets earn carbon credits, which they can sell to those that fall short of the requirements to comply.
- CCCs will be traded on national power exchanges. The Central Electricity Regulatory Commission will regulate trading activities within the ICM, while the Grid Controller of India will act as the registry operator, maintaining records of obligated entities and trading transactions.
- India's carbon trading scheme involves both a compliance market for certain industries and a voluntary offset market for other entities. Voluntary projects like afforestation, renewable energy, and clean cooking generate CCCs. These non-obligated entities can then sell the credits, creating a secondary market.
- The CCTS initially covers nine energy-intensive industrial sectors such as aluminium, cement, fertilisers, iron and steel, petroleum refineries, pulp and paper, and textiles, which account for 16% of India's total emissions. The power sector (which accounts for 40% of India's GHG emissions) is a glaring omission and may be included later.
- Although IT and electronics are not part of the initial list, this is an opportunity for tech companies and private investors to invest in carbon offsets in the voluntary market.